FYP-2 Mid Evaluation — Presenter Division and Speaking Scripts

Project S26-017-R-GASR — 4D Surgical Scene Reconstruction and Deformation Prediction

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Deck: **FYP2_Mid_Eval_Modules_5_and_6.pptx** — 17 slides · Total speaking time ≈ 19 minutes

1. How the Slides Are Divided

Each member presents one unbroken run of slides. The deck has been **reordered** so that the slides belonging to each member's subsystem sit together. That gives two handovers in the whole presentation, and nobody returns to the laptop after they have sat down.

Member	Slides	Time	The block they present
Ahmad Rashad 22i-1175	1 – 6 (6 slides)	≈ 6 min	Opens the talk and owns Module 5 end to end: the problem, what FYP-1 delivered, the model, every problem and fix inside it, and the architecture of the whole pipeline. The network, the diffusion formulation and the training system are his.
Shehroz 22i-0771	7 – 12 (6 slides)	≈ 7 min	Owns Module 6 and the entire evidence block: what the module does, its measurement problems, the training outcome, the baseline comparison, the key diagnostic finding and the uncertainty result. Every number in these six slides came out of his evaluation framework.
Humair Khan 22i-2623	13 – 17 (5 slides)	≈ 6 min	Owns the output and the close: the rendering and visualisation work that turns a prediction into something you can look at, then the literature positioning, the comparison against published work, what comes next, and the summary.

What changed in the deck to make this work

Three edits were needed. Without them, a sequential split would have put each of you on somebody else's slides.

- **The order changed.** The deck used to alternate between the two modules — module, module, problems, problems. It now runs *problem → Module 5 complete → Module 6 complete → output and close*, so each member's work is contiguous.
- **A slide was added for Humair (slide 13, "Making the prediction visible").** His subsystem — the rendering bridge from 512 points to 58,000 Gaussians, the re-verified renderer, the six-panel comparison, the magnified motion arrows and the forecast past the end of the video — previously appeared only as two rows on somebody else's slide. It is the most demonstrable work in the project and it now has its own slide.
- **Two rows on "Module 6 — what went wrong" were replaced.** The two visualisation rows moved to Humair's new slide. In their place are two baseline-correctness findings that are Shehroz's own work: the constant-acceleration baseline was computing velocity at the wrong time instant, and what the code called a Kalman filter was actually an alpha–beta filter. Both were found, fixed and verified.

Handover map — two handovers in total

After slide	From → To	Cue line
6 Architecture	Ahmad → Shehroz	"That bottom half is the evaluation side. Shehroz built it, and every number you are about to see came out of it. Shehroz."
12 Uncertainty	Shehroz → Humair	"That is where our numbers stand. Humair will show you what this actually looks like, and then where it sits against the published work."

Practical note: agree beforehand that the person handing over steps back from the laptop and the next person steps in. Do not pass the clicker mid-sentence. If you are presenting over a shared screen, pick one driver for the whole talk and the other two simply say "next slide."

2. Ahmad Rashad (22i-1175) — Slides 1 to 6

Opening, the problem, what FYP-1 delivered, Module 5, its problems and fixes, and the architecture of both modules. ≈ 6 minutes.

Slide 1 — From Reconstruction to Anticipation · 30 seconds

On screen: title, subtitle, the three names, supervisor, department.

Good morning. I am Ahmad Rashad, and with me are Shehroz and Humair. Our supervisor is Dr Akhtar Jamil.

This is our FYP-2 mid evaluation. In FYP-1 we built a system that reconstructs a surgical scene in 3D from a single laparoscopic camera. That work is complete. FYP-2 asks a harder question: can the same system tell you what the tissue is about to do *next*? That is Modules 5 and 6, and that is what this presentation is about.

We will be direct with you about the result, because the interesting part of this project is not a number we beat — it is what we found out when we did not.

Slide 2 — The laparoscope shows the present, never the next moment · 1 minute

On screen: text column on the left, two cards on the right, four stat tiles along the bottom.

Here is the clinical situation. A surgeon operates through one camera. That camera gives a flat image. There is no depth, no 3D structure, and crucially no indication of how the tissue will respond to the instrument that is about to touch it.

Reconstruction research has solved the first half of that problem very well. It has not addressed the second half at all. Every published surgical 4D reconstruction system is retrospective — it models deformation that has *already* been recorded.

Point at the stat tiles.

And these four numbers are why this is hard. Zero depth sensors — it is a single monocular camera. Zero deformation labels, because no surgeon annotates tissue motion frame by frame. Ten reconstructed videos, which is a small corpus. And the thing we are trying to predict moves one to two pixels per frame. That last number is the real difficulty: the signal we are chasing is smaller than the noise in most of what we are measuring.

Slide 3 — FYP-1 built the reconstruction, FYP-2 adds the prediction · 50 seconds

On screen: the six-module pipeline strip, four stat tiles. Keep this fast — it is a recap, not a result. Do not let it eat into Module 5.

One minute on what already exists, so you know what Modules 5 and 6 are built on top of.

Trace the pipeline boxes left to right.

Modules 1 to 4 are complete and validated from FYP-1. Frames and masks, then 4D Gaussian reconstruction, then trajectory extraction, then self-supervised training windows. Modules 5 and 6 — the two in red — are FYP-2, and are what the rest of this presentation covers.

The reconstruction reaches 30.10 dB PSNR and 0.904 SSIM from a single monocular camera, tracking roughly 58,000 Gaussians. From that we built 7,371 windows with no human annotation at any point in the chain. That last part matters — nothing downstream of here required a surgeon to label anything.

Slide 4 — Module 5: learning how tissue moves · 1 minute 20 seconds

On screen: Input / Model / Output cards, two gold cards, mint strip at the bottom.

So that is the gap. Module 5 is our attempt to close it.

Module 5 is a conditional diffusion model. It learns the deformation of 512 tracked tissue points, and it is trained entirely without clinical labels.

Walk the three cards left to right, roughly ten seconds each.

The input is three things: the last four frames of those 512 points, a number saying how far ahead we want to predict — one, two or three frames — and a patch of random noise. The model is 16.3 million parameters: a temporal transformer that looks along time for each point separately, then two-way cross-attention, and it denoises over twenty steps. The output is the predicted movement of each point — and because we can re-run it with fresh noise, we get several plausible futures from the same input rather than one guess.

Now the two gold cards. These are the two decisions the panel is most likely to challenge, so take them deliberately.

Two design decisions worth defending. First, why diffusion rather than direct regression. Soft tissue under contact is genuinely uncertain — there is no single correct future. A single regression estimate hides that. Sampling the model repeatedly gives us a spread of outcomes, and Shehroz will show you later that this spread is a usable confidence signal.

Second, why the horizon is an input. In our original formulation the same four frames appeared in training with targets one, two and three frames ahead — three different right answers for identical input. Without telling the model which one we wanted, the task is unlearnable no matter how large the network is. Embedding the horizon and injecting it into both branches is what made the objective well posed.

All of it trained on 7,371 windows across the ten videos, with no human annotation at any stage.

Slide 5 — Module 5: what went wrong, and what we did · 1 minute 20 seconds

On screen: five problem/solution rows. Do not read all ten boxes aloud — summarise the first three quickly and spend your time on the last one.

Five problems in Module 5, each one found by a measurement rather than a guess.

The task had no single right answer — fixed by conditioning on the horizon, which I have already described. The signal was smaller than the noise: tissue moves 0.004 units while diffusion adds noise at scale 1, so we rescale the target to unit variance and invert that before every reported metric, which means every number you see today is in physical units. The raw trajectories had almost no momentum — correlation between consecutive motions was 0.077, essentially noise — so we added per-video adaptive smoothing, and that correlation rose to 0.89.

Fourth, training made the model worse after epoch 60. Held-out error rose while training loss kept falling. We measured that rather than assumed it, shortened the schedule, raised weight decay, and we keep the best checkpoint rather than the last one.

Slow down here. This is the most important design change in the project.

And the fifth is the one I want you to remember. We found that a model which simply gave up — predicted zero displacement everywhere — scored exactly the same as doing nothing. The naive baseline *was* the degenerate solution. So we changed what the model predicts: instead of the displacement, it now predicts the *correction* to constant-velocity extrapolation. If it gives up now, it falls back to an estimator that is already 41% better than naive. We moved the floor.

Slide 6 — Architecture of both modules · 1 minute 10 seconds

On screen: two bands. Top = Module 5 training. Bottom = Module 6 inference and evaluation. The vertical arrow on the right is the checkpoint. This slide is also your handover.

This is both modules on one slide, and the important thing is the arrow on the right.

Trace the top band left to right with your hand.

The top band is Module 5, which is mine. Four frames of 512 points go into a shared point encoder, then a temporal transformer — three layers, eight heads — then cross-attention in both directions, then an output head that predicts velocity. The yellow box underneath is the conditioning: the diffusion step and the prediction horizon are both injected into the transformer. That horizon arrow is the fix I described two slides ago, and it is the single change that made the problem well posed.

The band emits exactly one thing: a trained checkpoint.

Follow the vertical arrow down to the lower band.

That checkpoint is the entire interface between the two halves of this project — one file plus an architecture descriptor, and nothing else. It is deliberately narrow, and it is what let the three of us work in parallel without blocking each other.

The bottom band consumes it and never trains. Trajectory replay behind a verification gate, a DDIM sampler running twenty steps, then three different ways of questioning the model, all compared against the same two baselines — and the whole thing exports metrics and figures for the comparative study.

Handover. Step back from the laptop as you say this.

That bottom half is the evaluation side. Shehroz built it, and every number you are about to see came out of it. Shehroz.

> **Ahmad — questions likely to come to you**
>
> - **"Why diffusion? Isn't this overkill for 512 points?"** — Two reasons. The uncertainty is real and clinically meaningful, and diffusion gives it to us for free by sampling; slide 12 shows that spread correlates with actual error at 0.69, which no deterministic model can produce. And the parameter count is driven by the transformer, not by the diffusion.
> - **"Why 20 sampling steps and not 500?"** — Training defines 500; DDIM lets us sample in 20 with negligible loss. Module 8 studies that trade-off explicitly.
> - **"You trained on 10 videos. Isn't that far too few?"** — Yes, and we say so on slide 16. The

overfitting we measured after epoch 60 is exactly what you would expect at this corpus size.

- **"Why predict velocity rather than noise?"** — At a displacement scale of 0.004, noise prediction is numerically badly conditioned. The v-parameterisation stays well conditioned across the whole schedule.
- **"Why is the architecture slide at the end of your section?"** — Because it is the bridge. It shows where my half stops and Shehroz's begins.

3. Shehroz (22i-0771) — Slides 7 to 12

Module 6 and the whole evidence block: what the module does, its measurement problems, the training outcome, the comparison, the key finding and the uncertainty result. ≈ 7 minutes.

Shehroz — the tone to hold. Your block contains the one result that does not flatter us. Do not apologise for it and do not bury it. Report it the way you measured it, then show that you found the cause. A panel trusts a team that reports a loss accurately far more than one that explains it away. Slides 11 and 12 are where you win this presentation back.

Slide 7 — Module 6: using the model, and measuring it honestly · 1 minute

On screen: three teal cards, two brick cards, mint strip.

Thank you, Ahmad. Module 6 is mine, and the first thing to say about it is that it trains nothing. It loads that checkpoint, asks it three different questions, and grades every answer.

Three teal cards, about ten seconds each.

Ask once: where will the tissue be one, two or three frames from now? That is the most reliable mode. Ask repeatedly: feed each answer back in and reach ten frames ahead — error compounds, and we measure how fast. Ask five times with different noise: five possible futures from one input, and how much they disagree becomes the confidence estimate.

The brick card on the left is the one to emphasise.

And everything is graded against two deliberately stupid rules. "Nothing moves" — the naive baseline. And "everything continues in a straight line" — constant velocity. Our position is simple: if a 16 million parameter model cannot beat those two, we say so. We do not change the metric until it looks better.

Six quality tests sit on top of that, covering accuracy, rollout, diversity, smoothness and cross-scene generalisation. The first one, QT0, is about the data rather than the model, and I want to spend a moment on it.

Slide 8 — Module 6: what went wrong, and what we did · 1 minute 10 seconds

On screen: five problem/solution rows. Lead with the first; it is the one that shows methodological seriousness. The

last two show the same rigour applied to the baselines.

Module 6's problems are different in kind from Module 5's. They are almost all about measurement integrity rather than modelling, which I think is the right emphasis for an evaluation module.

The first is the one I care most about. Rollout evaluation needs continuous trajectories, but Module 4 stores fixed windows. So we rebuild each video's trajectory by replaying the upstream processing — same random seed, same smoothing window, same normalisation. But a rebuild can silently diverge, and if it did, every number in this project would be wrong and we would never know. So QT0 cross-checks the rebuild against the windows Module 4 actually saved, and it runs as a *gate* — nothing is predicted until it passes.

Second, the metric was penalising us unfairly. Chamfer Distance is minimised by the average of the distribution, not by any single sample, so scoring one sample charged the model a penalty of roughly 40% for free. We now average several sampled trajectories before scoring.

Third, our diagnostics were measuring one easy video — taking the first N windows drew almost entirely from a single recording. Spreading the evaluation set evenly across all ten showed that our earlier scores had been about four times too optimistic. We would rather know that.

The last two rows are about the baselines. State them plainly — they show you held the comparison methods to the same standard as your own model.

And two of our own baselines were wrong, which we found and fixed. Our constant-acceleration baseline used a backward difference, which gives the velocity at t minus a half, not at t — errors up to 0.15 against displacements of 0.004. We derived it properly and verified the new form is exact on quadratic motion. And what our code called a Kalman filter was really an alpha–beta filter with fixed gains and no covariance at all. We replaced it with a genuine two-state constant-velocity Kalman filter carrying the full 2-by-2 covariance, and verified it recovers a known velocity to within 2.6 times ten to the minus six.

We mention those because a comparative study is only as honest as its baselines. If we had left them wrong, we would have beaten them unfairly.

Slide 9 — Training converged cleanly, and then overfitted · 1 minute 5 seconds

On screen: four stat tiles, two cards, one wide statement box at the bottom.

So what came out of training. Ahmad's training system ran 500 epochs on 16.3 million parameters. Training loss fell from 0.36 to 0.012, and the best held-out checkpoint was at about epoch 60.

What went right: smooth convergence, no instability, no divergence. And because Colab disconnects, multi-session training resumes exactly where it stopped — that mattered more than it sounds.

What went wrong: held-out error bottomed out near epoch 60 and rose after about 260 while training loss kept falling. That is overfitting, and we measured it directly rather than assuming it.

Pause before the bottom box. Let them read it.

And here is the number. Our best held-out Chamfer Distance is 0.01568. The naive baseline — predicting that nothing moves at all — is 0.01557. We are very slightly worse than doing nothing.

The model converged to predicting almost no motion, which is exactly what the naive baseline does. Understanding *why* is the rest of my section.

Slide 10 — A two-line formula beats our model · 1 minute 15 seconds

On screen: the comparison table, the brick card on the right, the wide box at the bottom.

This is the full comparison, on the complete held-out set across all ten videos.

Walk the table rows, not the individual cells.

Nothing moves, constant velocity, and our diffusion model, at one, two and three frames ahead. Read it honestly: our model sits essentially exactly on the naive baseline at every horizon. Constant velocity — two lines of code, no parameters at all — beats both of us, by 41% at one frame ahead. And we lose on all ten videos, not on some subset we could explain away.

Now the bottom box. This is where you show the work.

Before concluding anything, we tested two explanations that would have let us off the hook, and eliminated both.

The first: maybe error grows the further we get from the training frames — drift. We correlated error against distance from training data and got rho of plus 0.02 with a p-value of 0.61. There is no drift.

The second: maybe dropped frames during preprocessing corrupted the timing. We checked all 333 evaluation origins and found zero dropped frames.

Rollout is also stable — error grows with an exponent of 0.61, which is sub-linear, so the model is not diverging. Eliminating those two explanations by measurement is what let us find the actual cause, which is the next slide.

Slide 11 — The model knows which way tissue moves, not how far · 1 minute 35 seconds

On screen: a large teal statement, five cosine tiles, two cards. This is the most important slide in the deck — spend the time.

Chamfer Distance alone cannot tell the difference between a model that learned nothing and a model that learned the right direction but the wrong distance. Both score the same. So we measured them separately.

These five numbers are the agreement between the direction we predict and the direction the tissue actually moves — cosine similarity, where zero means no relationship at all.

Point at each tile as you say it.

At one frame ahead, plus 0.57. At two, plus 0.40. At three, plus 0.26. It decays, and by five frames it is at

zero and by ten it is negative — but at the horizons the model was actually trained for, it is strongly positive.

Let that land before the interpretation.

Here is why that matters. A model that had genuinely learned nothing would read 0.00 everywhere. It does not. Plus 0.57 combined with naive-level error can only mean one thing: the direction is right and the magnitude is wrong. And that is a far more tractable problem than "the model does not work."

Two fixes follow directly from that measurement, and both are already implemented. First, predict the correction to constant velocity rather than the displacement itself — Ahmad described that on slide 5 — which moves the fallback from the naive baseline to an estimator 41% stronger. Second, calibrate the magnitude at inference: we solve for a single scale factor that best matches predicted motion to observed motion. The mathematics there is worth one sentence — once you scale optimally, the remaining error is proportional to one minus the cosine squared, so *any* non-zero directional agreement produces an improvement over naive. We have 0.57.

Both are implemented. Evaluating them is the next run.

Slide 12 — The model knows when it should not be trusted · 1 minute 15 seconds

On screen: three stat tiles, a wide statement box, two cards. This is your closing slide — end strong.

And this is the result we are proudest of, and it is independent of the accuracy ranking entirely.

We sample five futures from the same input and measure how much they disagree. The correlation between that disagreement and the error the model actually makes is 0.69.

In plain terms: when the five sampled futures disagree with each other, the prediction really is worse. The model knows when it is in trouble. Uncertainty also rises smoothly with how far ahead we predict — 0.00011 at one frame, 0.00055 at ten, roughly five times.

Now the clinical argument. This is your strongest ground.

Why this matters more than the accuracy number: a system that is confidently wrong is unusable in an operating theatre. A system that can say "do not trust me here" is usable even when it is imperfect. And no method in our comparison provides this — not the naive baseline, not constant velocity, not a deterministic network. It is a direct consequence of the generative formulation.

Say the caveat yourself, before the panel raises it.

One caveat, stated carefully: this is a correlation between predicted spread and realised error. Some of it may reflect that busy moments are simply both harder and more variable. We say exactly that in the paper rather than claiming more than we measured.

Handover. Step back from the laptop.

That is where our numbers stand. Humair will show you what this actually looks like, and then where it sits against the published work.

Shehroz — questions likely to come to you

- **"So your model does not work?"** — It does not beat the baseline on Chamfer Distance, and we report that. It does learn motion direction at 0.57, and it does produce calibrated uncertainty at 0.69. Those are two measured positives and one measured negative, and we have localised the cause of the negative to magnitude.
- **"Why not just use constant velocity then?"** — For accuracy at one frame, today, you should. Constant velocity gives you no uncertainty, no multiple futures, and no mechanism to improve. Our point is that we now know exactly what to fix to pass it.
- **"Is 0.69 correlation actually good?"** — For an uncertainty estimate that was never trained against an error signal, yes. It is emergent from sampling, not fitted.
- **"How do you know your evaluation is not broken?"** — That is precisely what QT0 is for, and it is the first row on slide 8. The trajectory rebuild is verified against Module 4's saved windows before anything is predicted.
- **"Why Chamfer Distance?"** — It is correspondence-free, which matters because point identity is not guaranteed across frames. We also report Mean Point Error in the comparative study for the correspondence-aware view.
- **"You found bugs in your own baselines — how do we know there are no more?"** — We verify each baseline against a case with a known closed-form answer: the constant-acceleration form is exact on quadratic motion, and the Kalman filter recovers a known velocity to 2.6e−06. That is the standard we hold them to.

4. Humair Khan (22i-2623) — Slides 13 to 17

The visual output, the literature positioning, the comparison with published work, future improvements and the close. ≈ 6 minutes.

Slide 13 — Making the prediction visible · 1 minute 30 seconds

On screen: three teal cards, two gold cards, three PSNR stat tiles. This is your own subsystem — this is the slide where you have the most to say.

Thank you, Shehroz. Everything you have just heard is a number. My part of the project is turning those numbers into something you can actually look at, because a surgeon is not going to read a Chamfer Distance.

The difficulty is a mismatch. A prediction moves 512 tracked points. A picture needs all 58,000 Gaussians in the scene.

Three teal cards, left to right.

So the first thing I built is the bridge: distance-weighted interpolation that carries the predicted motion of

the nearest tracked points onto every Gaussian, weighted so that a tracked point receives its own predicted motion exactly, not an approximation of it.

Second, rendering that result meant re-implementing Module 2's renderer inside the prediction module — the positional encoding, the deformation network, and degree-3 spherical harmonic colour. I verified that re-implementation against the original rather than assuming the two matched, which is how I know the stored weights load correctly and the colour computation is identical.

Third, the output is six panels for every predicted moment: the real frame, the reconstruction ceiling, our prediction, the naive baseline, a motion overlay and an amplified error map — with image metrics computed over the full frame and over tissue regions alone.

Now the two gold cards. These are two problems that made the visual output misleading, and what I did about each.

Two honesty problems. Predicted motion was invisible — at one to two pixels, the predicted and real frames look identical to the eye, so you could show anything and claim it worked. Motion is now drawn as arrows coloured by agreement with the truth and magnified automatically, and the magnification factor is printed on every panel and never applied to any reported number.

And the second one is more serious. Nothing on screen was actually a prediction. Every predicted frame was being paired with a frame that already existed, so what we were showing was reconstruction, not anticipation. The forecast now runs twelve frames past the end of the recording — the real-video panel stops and says that no footage exists beyond that point, while the predicted panel keeps going. That is the only part of this project that shows prediction rather than describing it.

Point at the three stat tiles.

And in image space the numbers tell the same story Shehroz's geometric numbers did. The reconstruction ceiling is 30.04 dB — that is the best any prediction could possibly look. Our prediction is 26.79. The naive baseline is 26.74. We are marginally ahead there, and the gap is small for exactly the same reason: magnitude, not direction.

Slide 14 — Four adjacent fields, none occupies the cell we need · 1 minute 10 seconds

On screen: a five-row table, last row highlighted. Work the columns, not the rows — the argument is that no existing row has all four properties.

So where does that sit in the literature. I want to be precise about the gap rather than claim nobody has done anything nearby, because that would not be true.

Four lines of work are adjacent to ours. Surgical 4D reconstruction — EndoNeRF, LerPlane, EndoGaussian, Deform3DGS — produces exactly the 3D geometry we want, but retrospectively. It does not predict the future.

Surgical video anticipation does predict the future — TPG-VAE, and the recent SurgVista work — but in 2D pixel space, and conditioned on knowing the instrument's action in advance. There is no 3D geometry in

the output.

Point cloud forecasting predicts 3D and does it well, but on rigid outdoor LiDAR scenes — PointRNN, MoNet, PCPNet. Rigid bodies and deforming soft tissue are very different problems.

And medical point diffusion — Xie and colleagues this year, on liver motion — uses diffusion over 3D points on exactly our kind of tissue, but estimates the *current* state from an X-ray projection, not a future one.

Now the last row and the last column.

Our claim is the bottom row, and it is a claim about the combination, not about any single column. Future 3D tissue geometry, from monocular video, with no annotation, and with a calibrated confidence estimate. Note the last column in particular — none of the four lines above us reports uncertainty at all.

Slide 15 — How we compare with published work · 1 minute 20 seconds

On screen: left = reconstruction PSNR table plus a mint card; right = three cards on the prediction base papers.

Dr Jamil asked us to identify base papers and compare against them properly, so we have two families.

Left side first.

For reconstruction, the base paper family is LerPlane, EndoGaussian and Deform3DGS. They report 29.46, 30.25 and 30.48 dB. We report 30.10.

State the caveat immediately — do not let them find it.

That comparison is indicative, not controlled, and we say so in the paper: those published figures use *stereo* supervision on different sequences. What it does establish is that our reconstruction sits within 0.4 dB of the stereo state of the art while using a single camera and no depth sensor — and that geometry is the input Module 5 learns to predict.

Right side.

For prediction, two base papers. PointRNN, by Fan and Yang, is the closest published method to what we are doing — predicting future point clouds from past ones, scored with Chamfer Distance — and we have implemented it faithfully as our primary learned baseline. And Luo and Hu's CVPR 2021 paper is the first diffusion model over 3D point clouds; our denoiser is a conditional, temporally grounded instance of that idea.

The full comparison is eleven methods on one protocol — five classical, five learned, and ours. Module 7 is implemented; the run is the next step.

Slide 16 — What comes next · 1 minute

On screen: five numbered rows. The order is the message.

Five things, and the order matters — the first two are the ones that could change the headline result, and both are already written.

One: evaluate the two fixes Shehroz described. Calibration needs no retraining at all, and on the mathematics he showed you, it should move accuracy below naive.

Two: retrain on the constant-velocity residual, which raises the model's floor from the naive baseline to one that is 41% stronger.

Three: hold out an entire video rather than frames from videos we also train on — that removes the optimism of selecting and reporting on the same material.

Four: complete the eleven-method comparison. Modules 7 and 8 are implemented; the runs remain.

Five: grow the dataset beyond ten short recordings. The overfitting we measured is entirely consistent with a corpus this size.

Slide 17 — Where Modules 5 and 6 stand · 1 minute

On screen: dark closing slide. Delivered / Not yet achieved, and the closing line.

To close, plainly.

Delivered: a complete prediction pipeline that needs no clinical labels. A six-test evaluation framework with an integrity gate. Calibrated uncertainty at 0.69. A comparative study across eleven methods, implemented. And a conference paper drafted in IEEE format.

Not yet achieved: beating the naive baseline on accuracy. We are not going to present that as anything other than what it is.

But we did not leave it unexplained. We localised the cause to displacement magnitude rather than direction, and we implemented two fixes that follow directly from that measurement. Evaluating them is the immediate next step.

Pause. Deliver the last line slowly — it is the sentence you want them to remember.

An honest negative result with an identified cause is worth more than an unexplained positive one.

Thank you. We are happy to take questions.

> **Humair — questions likely to come to you**
>
> - **"Can we actually see a prediction?"** — Yes, and offer it. You have the six-panel side-by-side video and the forecast that runs twelve frames past the end of the footage. Have both open in another window before you start.
> - **"How do you know your re-implemented renderer matches the original?"** — Because it was checked rather than assumed: the stored weights load correctly and the spherical-harmonic colour computation is mathematically identical to Module 2's.
> - **"Isn't magnifying the motion misleading?"** — It would be if the factor were hidden or applied to the numbers. It is printed on every panel, and every reported metric uses unmagnified geometry.
> - **"Your PSNR is lower than Deform3DGS. Isn't that worse?"** — It is 0.38 dB lower, and they use stereo input on different sequences. We state that on the slide rather than presenting it as a

controlled win. The relevant point is that monocular reconstruction at this quality is good enough to serve as a prediction target.

- **"Has nobody really predicted future surgical geometry?"** — People have predicted future surgical *pixels*, with instrument actions as a condition. People have predicted future 3D for rigid outdoor scenes. People have used point diffusion on tissue for the current state. The combination in the last row of slide 14 is what we have not found in the literature.

5. Shared Question Routing

Agree this beforehand so you do not all start answering at once. Whoever is asked a question that belongs to someone else says one sentence and passes it: *"That is Shehroz's area — Shehroz?"*

If the question is about...	It goes to
The network, the diffusion formulation, why diffusion at all, training, overfitting, hyperparameters, the horizon conditioning, the residual target, the pipeline as a whole	**Ahmad**
Metrics, baselines, the quality tests, evaluation integrity, the comparison, the cosine finding, uncertainty, calibration, statistics	**Shehroz**
Rendering, the visual outputs, the forecast video, reconstruction quality, the literature, the base papers, the dataset, the runtime environment	**Humair**
Scope, timeline, division of work, what remains for FYP-2	**Ahmad**, then each member adds their own part

The one question to rehearse as a team. "Why should we pass a project that does not beat a two-line baseline?" Agree on a single answer and let one person give it. Suggested: *"Because we can tell you exactly why it does not, and the diagnosis is itself a result. We measured that the model learns motion direction at 0.57 and fails only on magnitude, we eliminated drift and dropped frames as explanations, and we implemented the two corrections that follow. We also produced a calibrated uncertainty estimate that none of the baselines can give you. We would rather bring you a measured negative with a cause than an unexplained positive."*